Diabetes Classification Report

In this assignment we used survey data to predict whether a person has prediabetes or diabetes. We followed the workflow from the course: understand the data, explore it, split it, prepare it, train a Logistic Regression, and evaluate it. The goal was not to get a high score. The goal was to understand each step and to read the results correctly. All the code and outputs are in our notebook.

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Q1. Target and class balance

The dataset has 253,680 people and 21 input features (X). The target (y) is Diabetes_binary. Class 0 means no diabetes. Class 1 means prediabetes or diabetes.

The classes are not balanced. We have 218,334 people in class 0 (86.1%) and only 35,346 in class 1 (13.9%). This matters because a model that always answers “no diabetes” would be right 86% of the time, but it would never find a single sick person. So a high accuracy can hide a bad model.

Q2. Understanding the features

We chose three features of different types:

- BMI is numerical. It is a real measurement, so a value of 30 means a BMI of 30.
- HighBP is binary. 0 means no high blood pressure and 1 means yes. It is an answer, not an amount.

- GenHlth is ordered categorical. It is the person's own rating of their health, from 1 (excellent) to 5 (poor). The order matters, but the gap between 1 and 2 is not necessarily the same as between 4 and 5.

In the CSV file everything is stored as a number, so the numbers alone do not tell us what a feature is. For example, Age = 9 does not mean 9 years old. It is a code for the group 60 to 64. We used the Understanding the Dataset guide to find the type of each feature: 3 numerical (BMI, MentHlth, PhysHlth), 4 ordered (GenHlth, Age, Education, Income) and 14 binary.

Q3. What we found in the EDA

We looked at all the features. For numerical features we used boxplots and compared the two classes. For binary and ordered features we compared the share of class 1 in each category. These five results were the most interesting:

1. General health. The share of class 1 goes from 2.5% (excellent health) to 37.9% (poor health). This was the clearest trend.
2. Age. It goes from 1.4% for ages 18 to 24 up to about 22% for ages 70 to 74, then drops a little (18.5%) for 80 and older.
3. High blood pressure. 24.4% of people with high blood pressure are in class 1, against 6.0% of people without it. Heart disease, stroke and difficulty walking show similar gaps.
4. BMI. The average BMI is 31.9 in class 1 and 27.8 in class 0. The two groups overlap a lot, so BMI alone cannot separate them.
5. A surprise. Heavy drinkers have a lower share of class 1 (5.8%) than other people (14.4%).

What we can conclude: these features are linked to the target in this dataset. What we cannot conclude: that one of them causes diabetes. An association is not a cause. The alcohol result shows why. One possible reason is that people change their habits after a diagnosis, but this is only an idea and EDA cannot test it. A few features, such as Sex, AnyHealthcare and Fruits, showed small differences. Weak alone does not mean useless, because a model can combine several weak signals.

Q4. Splitting and preparing the data

We used train_test_split with 80% for TRAIN and 20% for TEST, with stratify=y and random_state=42. TRAIN has 202,944 rows and TEST has 50,736. Class 1 is 13.93% in the full data, in TRAIN and in TEST, so stratifying worked. Without it, a random split could give us a TEST set with too few class 1 cases.

TEST must stay separate while we prepare the data and fit the model. TEST is like a final exam: it should show how the model does on people it has never seen. If the scaler or the model learns anything from TEST, information leaks from TEST into the model (data leakage) and the results look better than they really are.

Missing values. There are no missing values, so we did not need imputation.

Scaling and encoding. We chose the preparation according to the feature type:

- Numerical features (BMI, MentHlth, PhysHlth) are scaled with StandardScaler. It subtracts the TRAIN mean and divides by the TRAIN standard deviation. This is needed because BMI is around 28 while the health variables go from 0 to 30 days, and we want them on a comparable scale.
- Ordered features (GenHlth, Age, Education, Income) are encoded with OrdinalEncoder, with the categories given in their true order. We then scaled them too, which was our own choice.
- Binary features are already 0 and 1, so we kept them unchanged.

We did not need one-hot encoding, because this dataset has no unordered categories (like a list of chest pain types). Everything is done inside a ColumnTransformer and a Pipeline. The preprocessing is fitted on TRAIN only and then applied to TEST with transform.

Q5. Errors and metrics

We trained a Logistic Regression in the same Pipeline and predicted on TEST with a threshold of 0.50. The confusion matrix is: TN = 42,622, FP = 1,045, FN = 5,953, TP = 1,116.

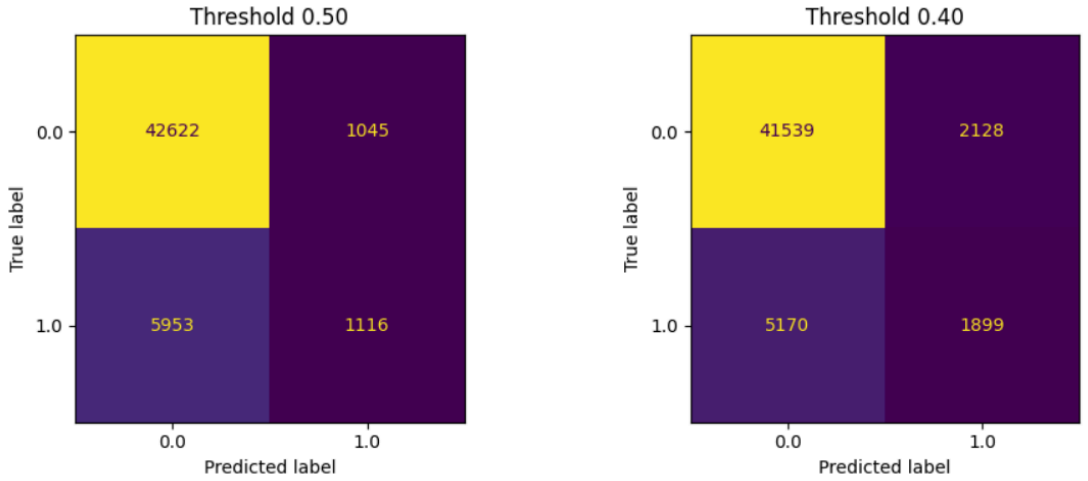
- A false positive is a person without diabetes who is predicted to have it. The result is a worry and maybe an extra medical test that was not needed.
- A false negative is a person with prediabetes or diabetes who is predicted to be healthy. The person is missed and may get help later. In a health problem this is usually the more serious mistake.

Accuracy is 0.862, but always predicting class 0 gives 0.861, so the model is barely better than doing nothing by this measure. Precision is 0.516: when the model says class 1, it is right about half of the time. Recall is 0.158: the model finds only 16% of the people who really are in class 1 (it misses 5,953 out of 7,069). F1 is 0.242. Accuracy alone is not enough because class 1 is rare, so the model can be right most of the time just by predicting class 0.

Q6. Probability and threshold

Logistic Regression gives a probability for class 1. The threshold is the rule that turns this probability into a class. We compared 0.50 and 0.40 using the same probabilities.

	Threshold	TN	FP	FN	TP	Accuracy	Precision	Recall	F1
0	0.5	42622	1045	5953	1116	0.862	0.516	0.158	0.242
1	0.4	41539	2128	5170	1899	0.856	0.472	0.269	0.342



With the lower threshold, the model predicts class 1 for more people (4,027 instead of 2,161). False negatives went down by 783 and true positives went up by the same number, so recall rose from 0.158 to 0.269. But false positives more than doubled, so precision fell from 0.516 to 0.472. Accuracy dropped slightly. This is a trade-off: we find more real cases, but we also raise more false alarms. For a health screening, missing a case is usually worse than an extra check, so 0.40 can be defended.

What did not change: the model, its coefficients, the preprocessing, the predicted probabilities, the TEST set and the true labels. Nothing was retrained. Only the decision rule changed.

Q7. Conclusion and limitations

From the EDA we learned that general health, age, blood pressure and income are clearly linked to diabetes in this survey, and that no single feature separates the two classes by itself. From the model we learned that a good-looking accuracy can be misleading when classes are imbalanced. At 0.50 the model misses most people with diabetes, and lowering the threshold helps recall but costs precision.

Limitations:

1. Association, not causation. The data come from one survey at one point in time, so we cannot say that any feature causes diabetes.
2. Imbalance. We did not use any method to handle the imbalance (such as class weights), so the model stays cautious and its recall is low.
3. Survey data. The answers are self-reported and can contain mistakes. The target also puts prediabetes and diabetes in the same class.
4. Simple setup. Logistic Regression is a linear model, and we used one single TRAIN and TEST split, so results could change a little with another split.
5. We compared the thresholds on TEST only to understand the trade-off. A real final threshold should be chosen without looking at TEST.